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REPRODUCIBLE · VALIDATED · FROZEN

First frozen unlearning study

IIT Mandi · 19 Sep 2026

DP-ForgetBench

When is explicit client unlearning redundant?

A privacy-aligned audit of federated learning.

THE HOOK

Does explicit client unlearning actually add value when client-level differential privacy already limits individual-client influence?

Group_2

DATA STATE

FROZEN

3 seeds · 100 clients

A deletion request creates a problem.

Federated learning keeps client data local, but their information still contributes to the global model.

CLIENTS

Each client holds its own data locally.

→

FEDERATED TRAINING

Round-by-round aggregation produces a global model.

→

GLOBAL MODEL

`M_all` — every client has influenced it.

→

DELETION REQUEST

Client `i` asks to be removed from the federation.

THREE CONCEPTUAL RESPONSES

OPTION 1

Do nothing. Keep M_DP.

Release or continue using the existing client-level DP model. Cheapest after the request, no extra computation or storage.

OPTION 2

Apply an unlearning procedure M_U.

Designed to remove the deleted client's contribution. May need historical updates, retained data, or additional compute.

OPTION 3

Retrain the federation without client i → M_R.

The natural reference for deletion. Most expensive; an honest upper bound of what "removal" can mean.

THE PRACTICAL TENSION

Exact retraining is expensive. Explicit unlearning can reduce utility or expose historical data. DP-only is almost free — but does it sufficiently suppress the deleted client?

Three problems with evaluating federated unlearning.

PROBLEM 1

Privacy-scope mismatch.

Many privacy discussions focus on *example-level* DP, while the deletion request concerns an *entire client dataset*.

THIS STUDY USES

Client-level central DP.

PROBLEM 2

"Forgotten accuracy" alone is insufficient.

A method can reduce deleted-client accuracy without behaving like the model that would have existed had that client never participated.

SO WE COMPARE AGAINST

M_R the exact retrained model without the deleted client

PROBLEM 3

Retraining itself is stochastic.

Two independent retrains without the deleted client do *not* necessarily produce identical models.

ESTIMATE THE BAND

M_R, M_R'

The natural retraining variability band as the reference, not a single deterministic target.

Is the difference caused by unlearning larger than the normal variation between independent retrains?

An empirical redundancy frontier.

DP-ForgetBench is a systematic evaluation framework — not another unlearning algorithm.

THE CORE QUESTION

Instead of asking *"Which unlearning algorithm is best?"*, ask *"Under what conditions is explicit unlearning necessary at all?"*

FOUR POSSIBLE OUTCOMES OF THE FRAMEWORK

OUTCOME

Redundant.

Explicit unlearning adds no meaningful additional privacy benefit over DP-only.

OUTCOME

Beneficial.

A measurable privacy benefit while remaining aligned with retraining.

OUTCOME

Harmful.

Substantial utility / cost degradation without demonstrated privacy improvement.

OUTCOME

Inconclusive.

The evidence is insufficient to distinguish the methods reliably — typically wide intervals, too few seeds, or regime not yet probed.

Characterization over the four-tuple:
$(\varepsilon, \alpha,$ deletion size, request sequence$)$

All four are research questions, not claims. The benchmark maps regimes into these four — empirical, not declarative.

What exactly are we trying to measure?

Six queries — each asks for an evidence direction rather than a method comparison.

RQ1 · REDUNDANCY

When does explicit unlearning add no measurable privacy improvement?

Compared against client-level DP-only at matched ε and matched data partitioning.

RQ2 · DEPENDENCE ON ε

How does redundancy change with the privacy budget ε?

Sweep ε from weak to strong privacy regimes. Where does the redundant / beneficial boundary sit?

RQ3 · HETEROGENEITY

Does heterogeneity α change the deletion picture?

RQ4 · DELETION SCOPE

One client vs. a larger fraction — does the answer shift?

RQ5 · REPEATED REQUESTS

Does the conclusion survive multiple / sequential deletions?

RQ6 · COST & EXPOSURE

Is a small privacy lift worth the compute, comm, & exposure cost?

Each query is operationalized as a numerical cell in `(ε, α, deletion size, request sequence)`. The framework answers them simultaneously, not in isolation.

We compare four candidate models.

BASELINE

M_DP

The original client-level DP global model.

No post-request action. Cheapest in compute, storage, exposure.

CANDIDATES

M_U

Explicit unlearning. Currently evaluated: retained-data fine-tuning · cached reconstruction · FedEraser-style procedure.

DELETION REFERENCE

M_R

Retrain the federation without the deleted client. The upper-bound honest reference for deletion.

VARIABILITY PROBE

M_R′

Another independent retrain without the deleted client. Estimates natural retraining variability.

THE QUANTITY THAT MATTERS

Not simply *"Are M_U and M_DP different?"* — but:

Does M_U provide additional privacy benefit beyond M_DP, while remaining consistent with M_R?

Privacy must match the deletion unit.

The DP mechanism is applied at the client level, not the example level.

TRAINING PIPELINE — CLIENT-LEVEL DP



ADJACENCY

Two datasets are neighboring if they differ by **one complete client's dataset.**

PRIVACY PARAMETERS

$\delta = 10^{-5}$

$\varepsilon \approx 32$

Current frozen study; delta fixed; ε tracked via a Poisson-sampled Gaussian RDP accountant.

IMPORTANT SCOPE STATEMENT

Secure aggregation is *not* implemented in the current study.

Methods that re-query retained raw data or rely on sensitive historical updates are treated as separate exposure regimes.

Privacy is matched to the deletion unit — clients, not examples. This is what makes DP a defensible baseline for client deletion.

CLIENT-LEVEL

Neighbors differ by one entire client's dataset.

EXAMPLE-LEVEL (NOT USED)

Neighbors differ by a single example. Wrong scope for client deletion.

The benchmark distinguishes three regimes.

Each regime maps to a different conclusion about whether unlearning is meaningful.

REGIME 1

Utility collapse.

A model performing near CIFAR-10 chance (~10%) cannot reveal much about any individual client. A low attack signal is meaningless when no useful signal exists.

REGIME 2

DP-dominated redundancy.

Model is useful; DP suppresses deleted-client influence. If M_DP sits inside the natural retraining band, unlearning may be redundant.

REGIME 3

Unlearning-sensitive.

Influence remains detectable beyond retraining variability. Here, explicit unlearning may provide a real benefit.

UTILITY VALIDITY GATE

Run-fraction gate introduced because early experiments produced "privacy" signals when the model itself had collapsed.

< 25%	**INVALID**	Discard the cell.
25–40%	**WARNING**	Record the caveat; do not anchor conclusions on these cells.
≥ 40%	**VALID**	Eligible to drive a redundancy claim.

A signal under collapse is not real signal. The gate is the difference between demo-grade "privacy illusion" and a sanity-checked redundancy claim.

Decision tree → regime 1 | regime 2 | regime 3 then gate → INVALID / WARNING / VALID

From exploratory experiments to the frozen study.

PHASE 1 — EXPLORATORY

Earlier binary CIFAR-10 grid.

GRID

30 runs · 6 (ε × α) cells
5 seeds per cell

SETTINGS

$\varepsilon \in \{2, 8, \infty\}$
$\alpha \in \{0.2, 0.75\}$

WHAT IT SUGGESTED

At ε = 2 and ε = 8, marginal JS benefit was about −0.001 to +0.013 over DP-only. Strongest-privacy regimes hinted DP could dominate the signal.

STRENGTHEN

not final

PHASE 2 — CURRENT FROZEN STUDY

A corrected, audited, reproducible pipeline.

WHAT WE ADDED

BACKEND	MODEL	STATISTICAL	AUDIT
Real 10-class CIFAR-10	GroupNorm CNN · ResNet path	Retrain ensembles + 3 seeds	Audited MIA + MUB

BASELINES	REPRODUCIBILITY
FedEraser-style procedure + positive unlearning controls	T4 / Colab execution path

OUTCOME

A first frozen multiclass unlearning study intended to support a *reproducible* redundancy claim — not the final word, but the first firm cell of the map.

First frozen unlearning study.

The configuration every later slide quotes by default.

FROZEN PARAMETERS

DATASET	CIFAR-10	10 classes
MODEL	SmallGroupNormCNN	
CLIENTS	100	200 / client
HETEROGENEITY	α = 0.5	
ROUNDS	40	5 local epochs
PARTICIPATION q	q = 0.5	
PRIVACY	ε ≈ 32	δ = 10^{-5}
SEEDS	20260918 · 20260919 · 20260920	

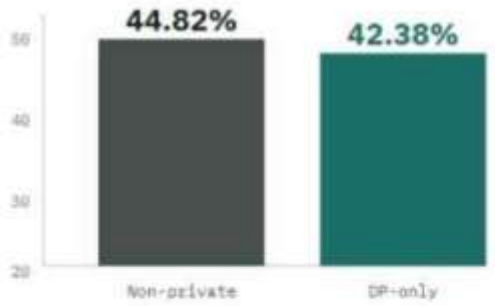
INDEPENDENT SEEDS · 3 RUNS

20260918, 20260919, 20260920

Three runs is sufficient to defend a frozen cell; not yet sufficient for narrow statistical claims.

UTILITY VALIDATION

Does the current DP configuration retain substantial utility?



RELATIVE DP UTILITY

≈ 94.6%

42.38 / 44.82. The current configuration passes the validity sanity check.

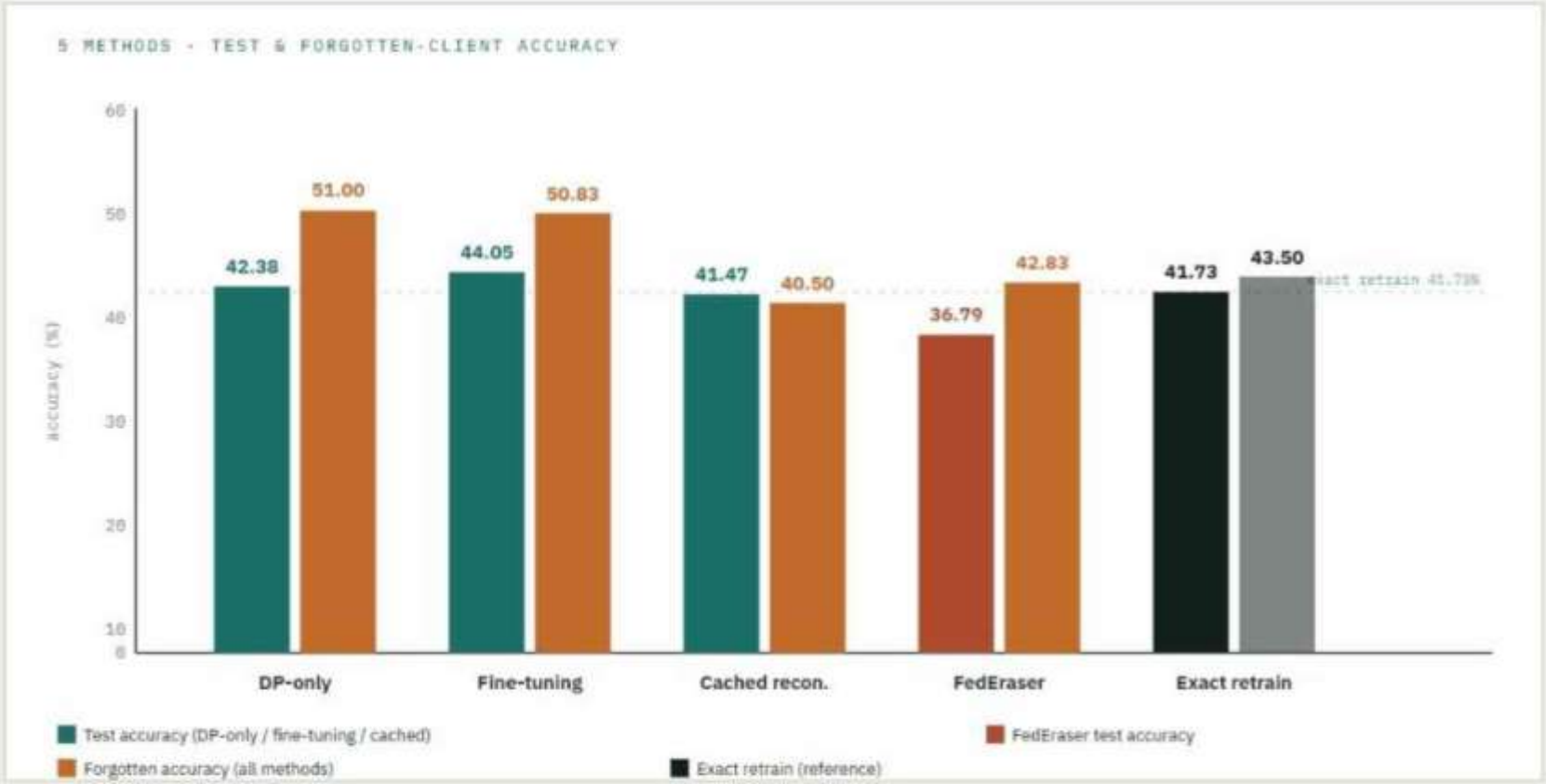
SANITY CHECK STATUS

The current DP configuration retains substantial utility and passes the validity sanity check.

Cell is eligible to drive a redundancy claim — slides 11–18 reuse these numbers.

What happens after the deletion request?

5 methods, 3 seeds · current 3-seed mean.



THE TIGHTEST GAP

0.65

percentage points

DP-only sits only **0.65 pp** above exact retraining in mean accuracy (42.38 vs 41.73).

FedEraser-style drops test accuracy to 36.79% — a substantial utility penalty without a demonstrated privacy advantage. Cached reconstruction moves forgotten accuracy to 40.50% but requires historical updates.

At this configuration, DP-only already behaves close to exact retraining in utility; explicit methods do not yet demonstrate a reliable additional privacy advantage.

A lower forgotten-client accuracy does not prove deletion.

Forgetting must be evaluated as a population-level behavioural change, never as a single drop in accuracy.

POPULATION 1

Forgotten.

Examples belonging to the deleted client.

Should the method actually be removing client-specific information here?

POPULATION 2

Retained.

Examples belonging to the clients that remain in training.

Utility preservation — does the model still serve the remaining federation?

POPULATION 3

Unseen.

Examples not used for training.

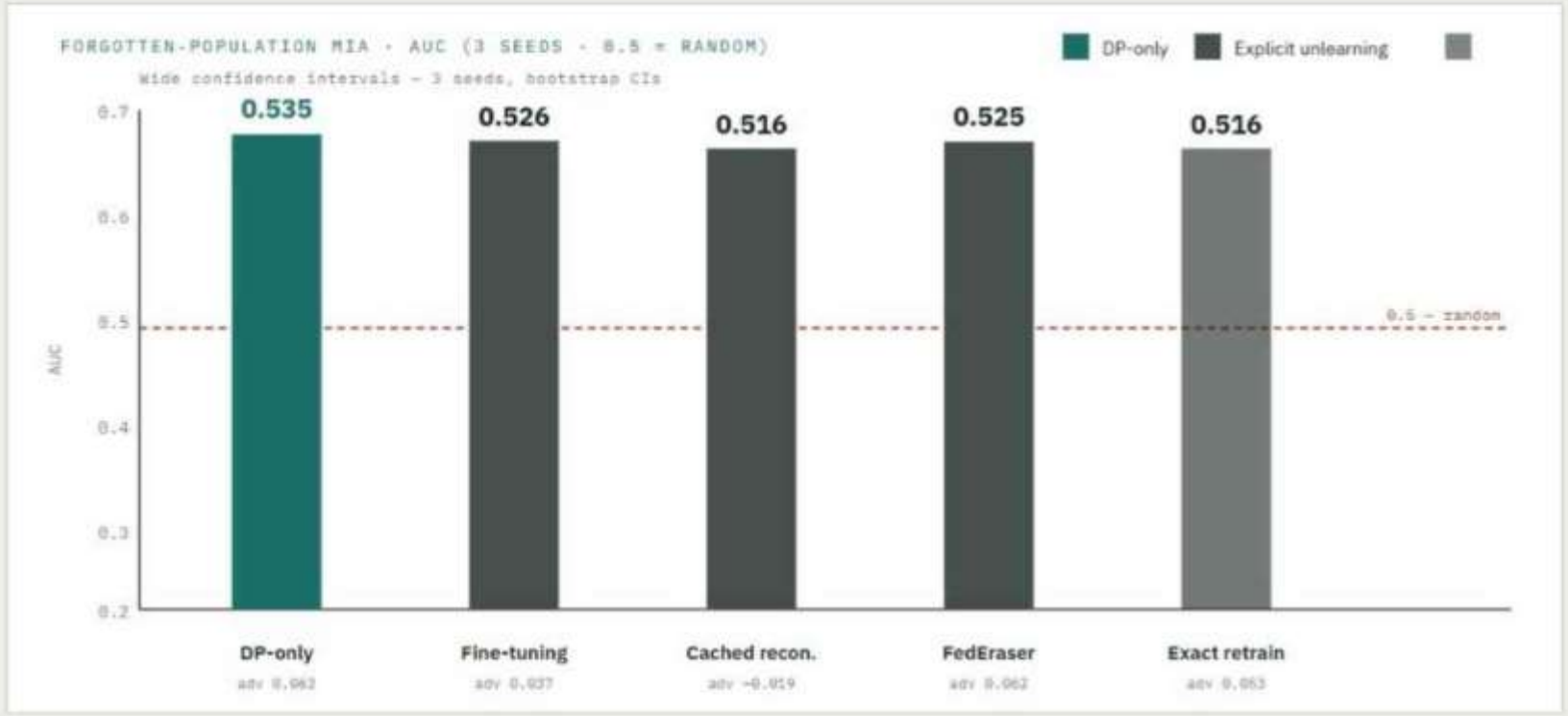
Generalisation — is the model still behaving like itself outside training?

A VALID UNLEARNING METHOD

should ideally reduce deleted-client information, remain aligned with the exact-retraining reference, preserve retained-client utility, and preserve generalisation to unseen data.

Auditing privacy rather than assuming it.

A label-matched membership attack with separate calibration and held-out evaluation, threshold picked on calibration only.



ATTACK DESIGN

Label-matched members

Unseen examples

Split — separate calibration & eval

CALIBRATION
pick threshold

HELD-OUT
report AUC + adv

REPORTED

AUC · advantage · bootstrap 95% confidence intervals over 3 seeds.

DP-only AUC is close to random (0.535 ≈ 0.5). Some methods reduce the score — but with only 3 seeds and wide intervals, there is ***no statistically reliable evidence*** *that any explicit method improves privacy over DP-only.*

Important distinction: the method must also be evaluated relative to retraining variability, utility, and lifecycle cost.

(full sentence continues right)

LOWER ATTACK SCORE

does not automatically mean better unlearning.

The benefit of being explicit, per metric.

DEFINITION

`MUB = Metric_U − Metric_DP` · depends on the metric and its direction.

BE CAREFUL ABOUT THE SIGN

Negative MUB is not automatically better. Direction depends on the metric.

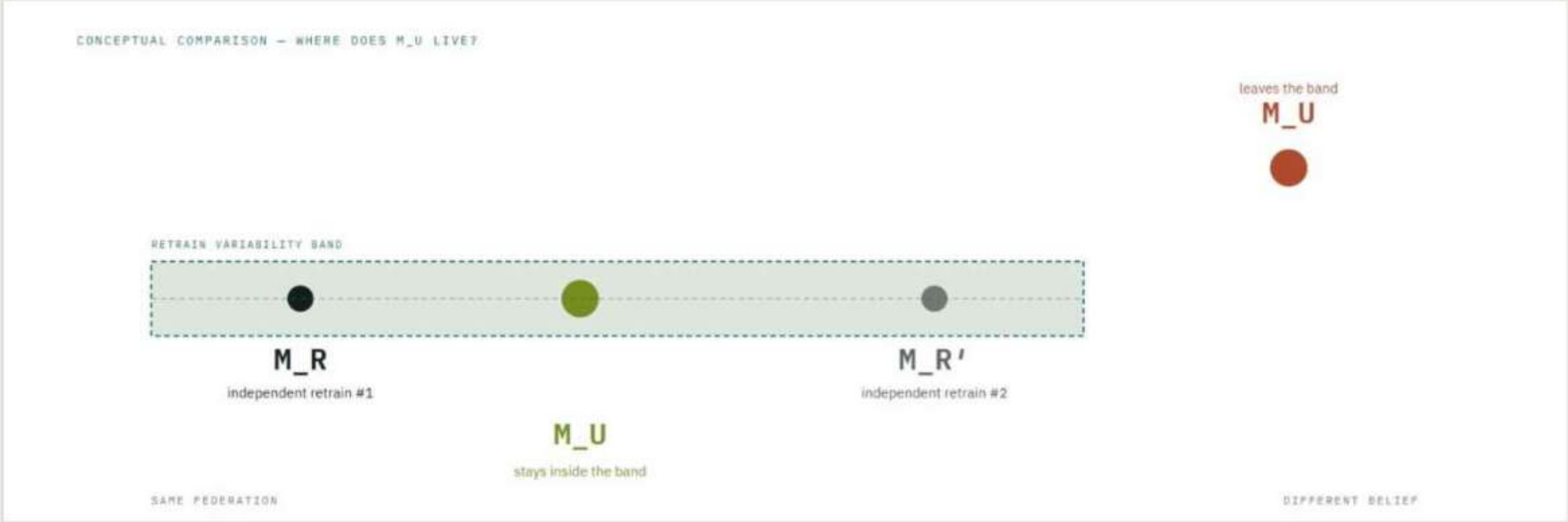
MUB PER METHOD × METRIC

Method	Accuracy MUB	JS MUB	MIA MUB	ATTACHMENT
Fine-tuning RETAINED-DATA POST-REQUEST	+0.0167 utility lift	+0.0061 small drift	−0.0026 near zero	*looks like a utility adaptation, not demonstrated forgetting*
Cached reconstruction HISTORICAL UPDATE ACCESS	−0.0092 utility drop	−0.0077 misaligned with retrain	−0.0060 but needs sensitive history	*trades utility + cost for a small attack reduction*
FedEraser-style HISTORICAL COMMUNICATION	−0.0559 largest utility penalty	−0.0027 near zero	−0.0226 wide-interval	*largest utility loss with no demonstrated privacy gain*

The MUB table is a per-metric delta. A method reads as beneficial only when MUB is positive on the right metric, not costly on the others, and within retraining variability.

Exact retraining is not a single fixed target.

Stochastic optimisation means M_R and $M_{R'}$ are not numerically identical.



Does explicit unlearning move the model **outside** the variability that naturally occurs when we retrain without the deleted client?

An unlearning method is not free.

A small privacy improvement may not justify substantial additional lifecycle cost or exposure.

METHOD

DP-only.

POST-REQUEST

nothing

no post-request training · no additional communication · no historical-update requirement

The cheapest option once the request lands.

METHOD

Fine-tuning.

POST-REQUEST

retained-data access

Continues training on the remaining federation. Assumes raw retained data is still available.

Treated as a separate exposure regime — not free DP post-processing.

METHOD

Cached recon.

HISTORICAL STORAGE

≈ 497 MB

per compact model/run

Stores historical per-round / per-client updates so the model can be retroactively corrected after deletion.

METHOD

FedEraser-style.

HISTORICAL COMMUNICATION

≈ 965 MB

current communication estimate

Reassembles historical client contributions from cached rounds — large storage *and* communication footprint.

ACCOUNTING RULE

Do not charge DP-only with historical storage merely because other baselines collect shared history. Charge each method only for the resources it actually requires.

What the current frozen study actually shows.

EVALUATED AT

$\varepsilon \approx 32$ · $\alpha = 0.5$ · 40 rounds · 5 local epochs · 3 seeds

BASELINE

DP-only.

TEST ACC	FORGOTTEN
42.38%	51.00%

MIA AUC 0.535 · advantage 0.062 · no post-request action

EXPLICIT

Fine-tuning.

TEST ACC	FORGOTTEN
44.05%	50.83%

Improves accuracy but does not clearly reduce forgotten-client leakage.

EXPLICIT

Cached recon.

TRADE

Reduces forgotten accuracy and MIA — but reduces utility and requires historical sensitive information.

EXPLICIT

FedEraser-style.

PENALTY

Reduces test accuracy to 36.79% — no demonstrated privacy advantage over DP-only.

REFERENCE

Exact retrain.

GAP

41.73% accuracy. DP-only is only 0.65 pp higher in mean test accuracy.

DP-only is close to exact retraining in utility. No explicit method yet demonstrates a reliable additional privacy advantage at this configuration.

What does the evidence currently suggest?

STRONGEST DEFENSIBLE STATEMENT

At the current frozen configuration, client-level DP provides a strong baseline against which explicit unlearning must demonstrate additional value.

Suggestive · but inconclusive.

THREE SHORT STATEMENTS

FINE-TUNING

Improves utility, does not clearly improve forgotten-client privacy.

CACHED RECONSTRUCTION

Lowers attack scores but sacrifices utility & alignment and requires historical-update storage (~497 MB).

FEDERASER-STYLE

Substantial utility penalty (~36.79% acc) without a demonstrated privacy advantage.

AGAINST EXACT RETRAINING

DP-only is only 0.65 percentage points above exact retraining in mean accuracy — but uncovers no statistically reliable additional privacy benefit at this configuration.

This is not yet sufficient to claim a universal redundancy frontier. It is, however, the first firm cell of a map intended to be filled by later work — see slides 21 and follow on.

What we are not claiming.

NOT A CLAIM

"DP perfectly deletes a client."

Client-level DP bounds information leakage. It is not being presented as exact deletion.

NOT A CLAIM

"DP always makes unlearning unnecessary."

The answer may depend on ε, heterogeneity, deletion size, client influence, request sequence, and the underlying model / dataset.

NOT A CLAIM

"There is a universal ε threshold."

The current $\varepsilon = 32$ result applies to the evaluated configuration. Other configurations remain to be probed.

NOT A CLAIM

"We have certified / legal deletion."

This is an empirical benchmark, not a legal certification. The outputs are research artefacts, not compliance statements.

NOT A CLAIM

"Unlearning is useless."

Some regimes may still benefit from explicit unlearning. The benchmark exists *to find them*, not to dismiss the discipline.

Correct framing: the goal is to **empirically identify when explicit unlearning is redundant, beneficial, harmful, or inconclusive** under a matched client-level DP threat model.

What remains to be validated.

01

Dataset scope.

The frozen study uses CIFAR-10 — not yet a universal federated-learning benchmark.

02

Number of seeds.

3 independent seeds.

03

Wide MIA intervals.

Confidence intervals remain wide — bootstrap CIs barely distinguish methods.

04

Tri-population audit.

Baseline logistic probe — not yet a full reproduction of TC-UMIA.

05

Secure aggregation.

Not currently implemented. Should be evaluated as a separate regime in the next stage.

06

Post-request data access.

Fine-tuning / FedEraser access retained raw data or historical info — treated as a different exposure regime.

07

Stronger baselines.

Published federated-unlearning baselines need to be reproduced under a matched threat model.

WHAT THE LIMITATIONS MEAN

Together, these limit the strength of statistical claims and prevent us from declaring a universal redundancy frontier. They are an outline for the next stage — see slide 21.

Eight steps to turn evidence into a defensible result.

STEP 01

Complete tri-population audit.

Forgotten / retained / unseen accuracies, macro AUC, confusion matrix, CIs.

STEP 02

Correct lifecycle accounting.

Charge each method only for the resources it actually requires (slide 16 rule).

STEP 03

10 seeds per cell.

Repeat transition & harmful cells in particular until intervals separate.

STEP 04

Heterogeneity sweep.

Multiple α values — does client heterogeneity change redundancy behaviour?

STEP 05

Deletion-size sweep.

Single-client deletion → larger deletion fractions.

STEP 06

Sequential deletion.

Partial · repeated · sequential client deletion requests.

STEP 07

Stronger unlearning baselines.

Reproduce published FU methods under the same privacy / threat model.

STEP 08

Generalisation beyond CIFAR-10.

Move to naturally client-partitioned datasets and broader heterogeneity.

Each step is operationalized in the same threat model and the same validity gate. Steps 1–3 tighten ***the current cell****; steps 4–6 expand the map; steps 7–8 push outward to publication-grade claims.*

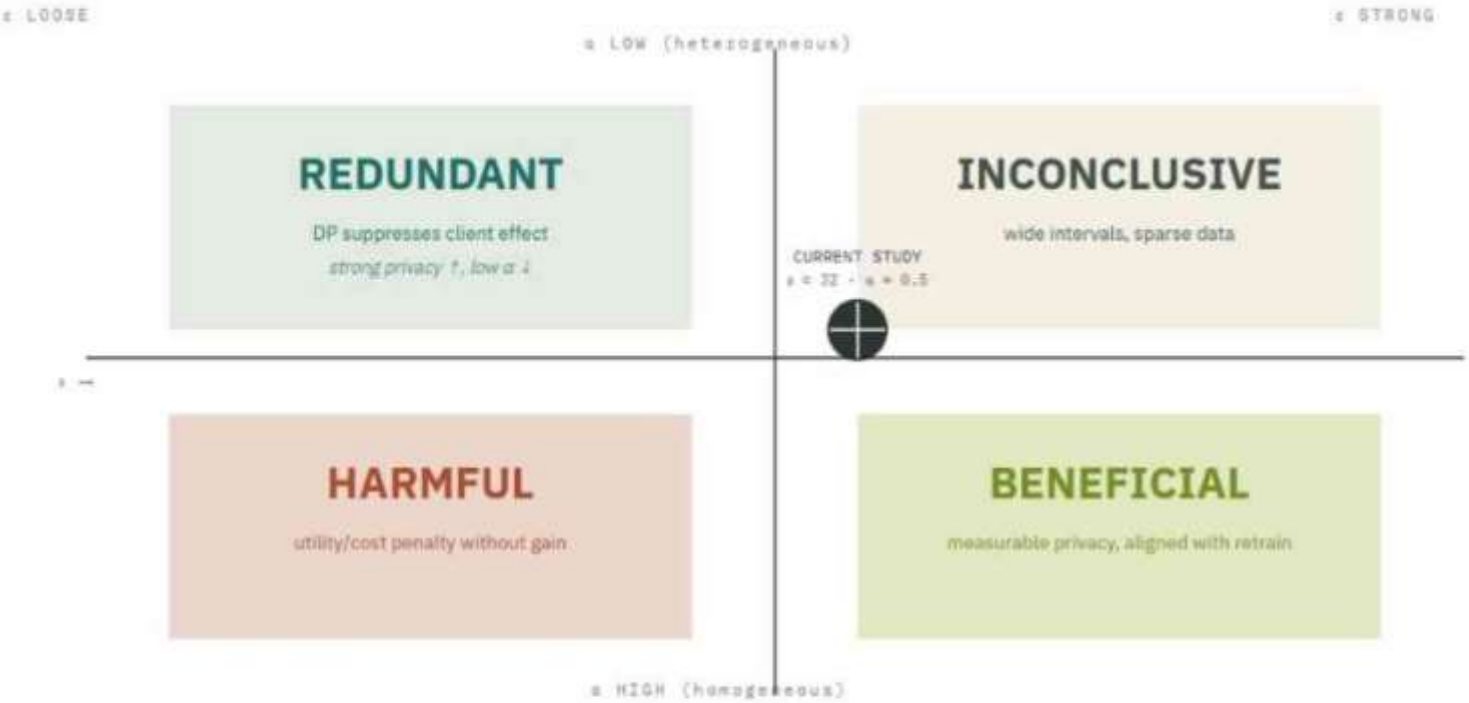
A redundancy frontier.

TARGET MAP

(ε, α, deletion size, request sequence)

NOTE

Conceptual target — *not* a completed empirical map.



PURPOSEFULLY RESTATED

REDUNDANT

DP already suppresses client effect.

BENEFICIAL

Privacy lift, aligned with retraining.

HARMFUL

Utility / cost penalty, no gain.

INCONCLUSIVE

Evidence not yet decisive.

Purpose: identify when explicit unlearning actually adds measurable value — not to crown one method universally superior.

This is the eventual shape of the benchmark. Today's study is one cell of it — not the map.

TRADITIONAL QUESTION

"Which federated unlearning algorithm should we use?"

DP-FORGETBENCH ASKS

"When do we need explicit client unlearning at all?"

THE BENCHMARK CONSIDERS

THREAD client-level privacy	MEASURE model utility	POPULATIONS forgotten / retained / unseen	AUDIT membership inference · retraining variability
SWEEP heterogeneity · deletion scope · request sequence	ACCOUNT FOR storage · communication · post-request data exposure	CURRENT STATUS **suggestive · inconclusive**	

FINAL RESEARCH GOAL

Determine when explicit unlearning is **genuinely necessary** — rather than assuming every deletion request requires an expensive procedure.

Thank you.
Questions, please.

POSSIBLE DIRECTIONS

QUESTION 01

How should redundancy be defined statistically?

Effect-size vs. interval separation; against what reference?

QUESTION 02

How does client heterogeneity affect DP masking?

When do central-tendency clients vs. outliers behave differently?

QUESTION 03

When does unlearning actually become beneficial?

Which parameter regions, if any, push M_U out of the retraining band?

QUESTION 04

How should lifecycle privacy exposure be accounted for?

Storage, communication, and post-request access as privacy-relevant quantities.

CONTACT

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STATUS

Reproducible · first frozen · 3 seeds

Faculty Research Review · September 2026

DECK TAG

`dp-forgetbench.slides`

24 slides · 1920×1080